DYNAMIC MODELLING AND SENSITIVITY ANALYSIS OF THE HOMOLOGOUS RECOMBINATION DNA REPAIR NETWORK

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Homologous recombination (HR) is the main pathway responsible for the repair of DNA double-strand breaks, which threatens genome stability. In this project, we developed and a dynamic computational model of the HR repair network to study both the structural organization and the dynamic response to perturbations. Starting from a HR pathway's network-based representation, we implemented a framework that combines graph analysis, ordinary differential equations (ODE) modelling and sensitivity analysis. The baseline network was expanded by introducing explicit repair-state variables and additional regulators involved in end resection, mediator exchange, and late repair stages, including CtIP, EXO1, DNA2, DSS1, RAD54, and FANCM. Especially, we improved the modelling of RAD51 loading and filament formation, due to its importance to strand invasion and repair completion. The model was analysed using local sensitivity analysis Morris screening, and Latin Hypercube Sampling with Partial Rank Correlation Coefficient analysis. The results demonstrated that the most influential mechanisms are associated with RPA-coated-single-stranded DNA, BRCA2-DSS1-mediated RAD51 exchange, RAD51 filament assembly, and FANCM-supported repair progression. Overall, the study shows that a dynamic system can provide a biological interpretation of the HR regulation and can identify the molecular processes that mainly control repair efficiency.

1.INTRODUCTION

DNA is constantly exposed to exogeneous and endogenous sources of damage. Among all, double-strand breaks (DSBs) are the most dangerous because both DNA strands are disrupted simultaneously. If not correctly repaired, DSBs can lead to chromosomal rearrangements, genomic instability, and cancer development.

To preserve genomic integrity, one of the most accurate mechanisms is homologous recombination (HR), which repairs DSBs by using an intact homologous DNA sequence as template. This process is especially active during the S and G2 phases of the cell cycle, so when a sister chromatid is available for error-free repair.

The HR pathway involves different coordinated steps. Firstly,the damage is detected by the MRN complex, which helps to end the resection and to generate the single-stranded DNA (ssDNA). Then, the exposed ssDNA is coated by the replication protein A (RPA), which stabilizes it. Subsequently, BRCA2 promotes the replacement of RPA by RAD51, allowing the formation of the RAD51 nucleoprotein filament required for homology search and strand invasion[1],[2]. The invaded DNA form a D-loop intermediate, that is processed by down streamed proteins (i.e. RAD52 and RAD54) until repair is completed. The paper by Lecca and Ihekwaba-Ndibe presents this biological sequence as the foundation for a mathematical framework in which the HR pathway is translated into a dynamic model, [3].

For DNA repair, graph representations allow the identification of topologically important proteins and dynamic models enable to investigate how perturbations propagate through the system over time. The cited paper proposes an automated pipeline that converts a network of HR interactions into a set of first-order differential equations and then applies parametric sensitivity analysis to identify the proteins and interactions that most influence system behaviour.

The aim of the present project was to build on this framework and generate a more mechanistic and interpretable HR model. Indeed, the original network-based model was improved by adding explicit repair-state variables and new biologically relevant factors involved in end resection and RAD51 regulation. The project focused especially on refining the RAD51 loading and exchange step, since the literature and the original study both indicate that RAD51 dynamics are central to HR but may be oversimplified in reduced models. Additional regulators such as CtIP, EXO1, DNA2, DSS1, RAD54, and FANCM were introduced in order to better represent the biological progression from initial break recognition to final repair. The overall objective was not only to simulate HR dynamics, but also to identify which molecules and parameters most strongly govern repair progression(see Figure 1,7).

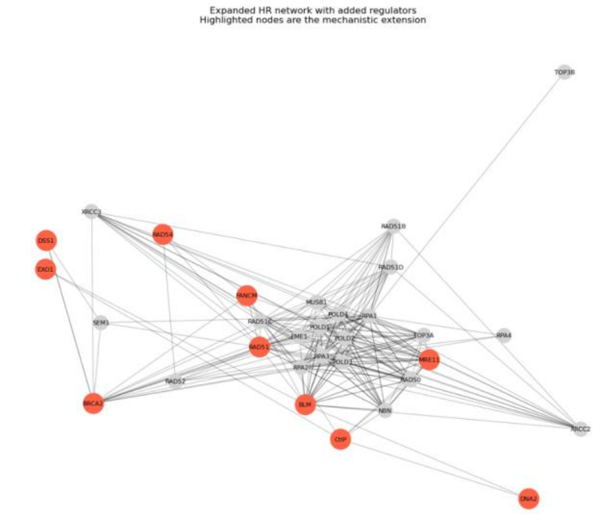

Figure 1 | Expanded HR network with added regulators. Highlighted nodes indicate the mechanistic extension introduced, including factors involved in end resection, RAD51 exchange and late repair progression.

2.METHODS

The starting point was the HR interaction network described in the literature with the same dataset used in the reference study, [4], where the HR pathway is represented as a graph of molecular and protein interactions. It contains 25 nodes and 250 edges, and it is used both for topological analysis and for the automatic generation of ODE-based dynamics,[4].

For this study, the baseline network was reconstructed and represented as a directed interaction graph. Degree, betweenness, clustering coefficient, eigenvector centrality and vibrational centrality were analyzed in order to retrieve the structural importance of nodes. The original paper emphasizes that these measures are used for identifying proteins that act as hubs, communication bridges, or vulnerable points in the HR system. Also, vibrational centrality was interpreted as a measure of node sensitivity to perturbation.

The dynamic part of the project was based on an ODE framework. Molecular interactions were converted into rate-based equations to express the abundance or activity of each component over time in terms of consumption, binding and transformation processes. New state variables were introduces to represent the main stages of the HR repair pathway (i.e. the initial DSB state, resected ssDNA, RPA-coated ssDNA, RAD51-loaded DNA, RAD51 filament formation, D-loop generation, and final repaired DNA) allowing the model to describe both the interactions between proteins and how the repair process evolves through distinct mechanistic phases.

A methodological extension was the refinement of RAD51 loading. Indeed, biological HR process, the exchange of RPA for RAD51 on ssDNA is mediated by BRCA2, with DSS1 acting as a cofactor that supports the displacement of RPA and facilitates RAD51 filament formation. The reference paper,[3], explicitly highlights the role of BRCA2-DSS1 in RAD51-RPA switching and in the assembly of the RAD51 filament. So, to capture this stage in a more realistic way, the network included explicit kinetic processes for BRCA2-DSS1 complex formation, BRCA2-RAD51 binding, RAD51 loading onto ssDNA, filament assembly, and filament disassembly or recycling.

Additional proteins involved in end resection and late HR steps were also included (i.e. CtIP, EXO1, and DNA2) to improve the representation of DNA end processing. RAD54 was included to account for post-invasion processing and for branch migration. Also, FANCM was included because of its known relevance in genome stability and also because prior analyses showed that extending the HR network with FANCM can alter centrality rankings and pathway sensitivity.

Then, the model was simulated over a defined time interval. Since most HR interactions' kinetic parameters are not available experimentally, the simulations were considered in arbitrary units rather than as quantitatively calibrated in vivo kinetics. The reference paper,[3], state the same, noting that both protein concentrations and rate constants are usually unknown and so treated within plausible parameter ranges.

To analyze which parameters are the most influential, three distinct levels of sensitivity analysis were performed. Firstly, local sensitivity analysis quantified the effect of small parameter perturbations on model outputs. Secondly, Morris screening was used to identify parameters with strong global influence and potential nonlinear interaction effects. Lastly, Latin Hypercube (LHS) with Partial Rank Correlation Coefficient (PRCC) analysis was used to understand how variations in parameters correlate with final repair outcomes (i.e. the level of repaired DNA). So, the overall sensitivity analysis provided information about the local influence near the baseline, global importance across the parameter space and the direction of association with the repair efficiency.

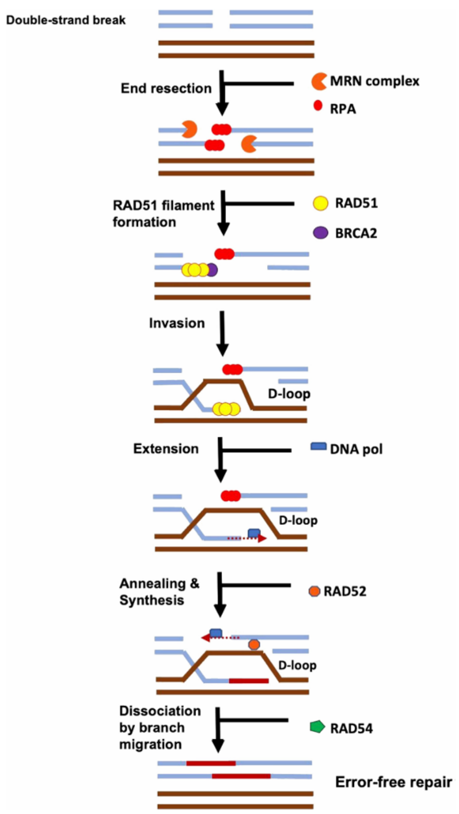

Figure 2 | Main steps of homologous recombination repair, from double-strand break recognition and end resection to RAD51 filament formation, strand invasion, D-loop processing, and final error-free repair.

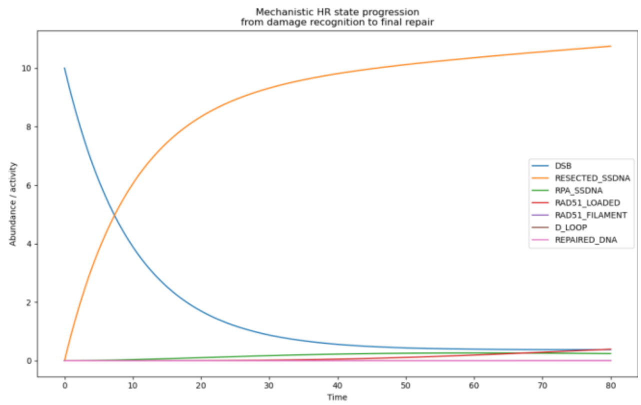

Figure 3 | Simulated temporal progression of mechanistic HR repair states.

3.RESULTS

The baseline HR network recreated the structure of the published HR network and provided the baseline for a first dynamic model. The centrality analysis demonstrated the expected importance of proteins associated with ssDNA processing and repair coordination, agreeing with the literature. In the reference study,[3], proteins like BLM, RAD50, RAD52, and the RPA complex resulted as structurally significant according to different centrality metrics, reflecting their roles as hubs, and bridge nodes.

The first major result of the study was that adding explicit repair-state variables produced a more interpretable dynamic model. Indeed, the simulation can capture the temporal progression of the repair process and can describe also the changes in protein abundances. The trajectories showed that DSB state decreased over time, resected ssDNA and RPA-bound states appeared early, RAD51-loaded and filament states emerged afterward, D-loop formation followed and repairs DNA accumulated at later stages, which is biologically coherent with the literature (see Figures 2,3).

The second major result was the improved representation of RAD51 dynamics. Indeed, after refining the BRCA2-DSS1-RAD51 exchange module, RAD51-related sensitivity increase of approximately two-fold . This suggests that the baseline model in the reference study, [3], underrepresented the control exerted by RAD51 loading and filament formation. This result is particularly meaningful since the reference article itself, [3], notes that unexpectedly low RAD51 sensitivity, may arise from oversimplified modelling assumptions.

The constructed network also changed the relative importance of specific regulators, since the local sensitivity analysis showed that the most influential parameters were associated with RPA binding to ssDNA, BRCA2-DSS1 complex formation, BRCA2-RAD51 association, RAD51 loading and RAD51 filament assembly. These parameters affected multiple features of the model such as intermediate repair states and finale repaired DNA. This reflects that the transition from RPA-coated ssDNA to a productive RAD51 filament is one of the main limitation of the HR process in the model (see Figure 4).

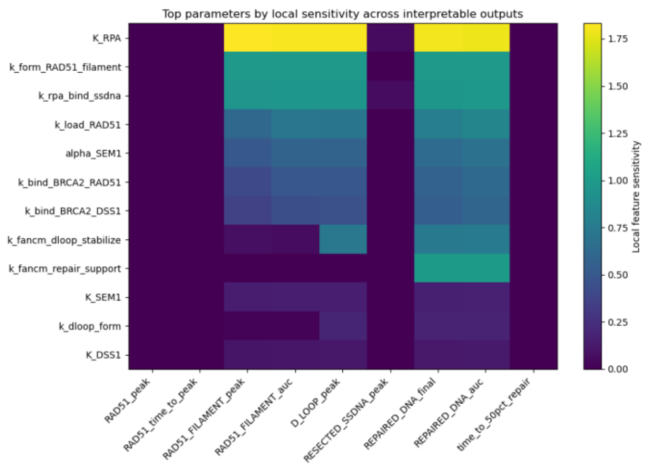

Figure 4 | Local sensitivity heatmap of the mechanistic HR model.

The Morris screening analysis confirmed what written above from a global perspective. Parameters controlling RAD51 filament assembly and BRCA2-mediated loading showed high overall influence, signs of interaction and nonlinearity. Also, FANCM-related parameters ranked highly in the global analysis, especially the ones linked to D-loop stabilization and downstream repair support, suggesting that FANCM contributes substantially to the global control of repair progression in the expanded model. This is consistent with the reference study,[3], which found that incorporating the FANCM pathway modifies the centrality and sensitivity structure if the HR network and identifies FANCM as a highly connected and influential node in the improved network (see Figure 5).

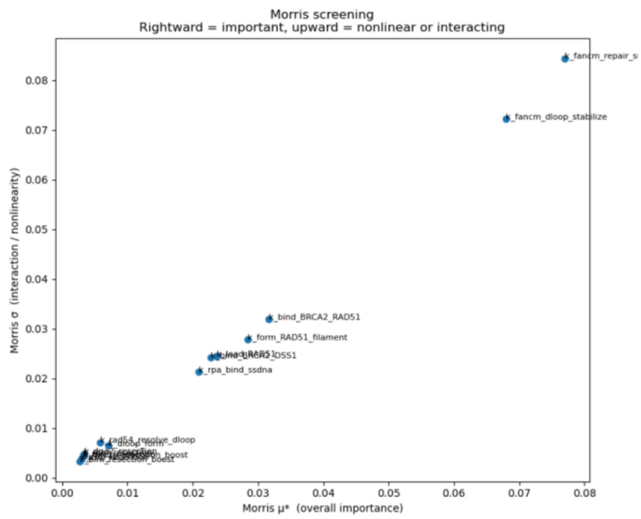

Figure 5 | Morris global sensitivity analysis of the mechanistic HR model.

The LHS-PRCC analysis provided directional information about parameter effects. The ones linked to productive HR progression, like RAD51 loading and filament formation, were positively associated with the final repaired DNA outcome. On the contrary, parameters that limited these transitions showed weaker or negative associations. Indeed, the scatter plots generated from the sampled parameter space further supported these PRCC trends, since they showed monotonic relationships between main control parameters and repair completion (see Figure 6).

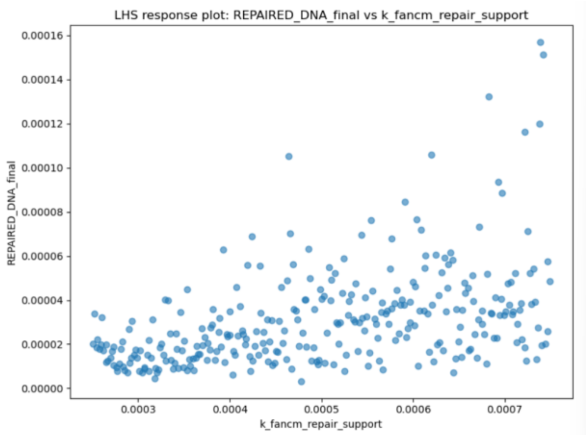

Figure 6 | LHS response plot relating final repaired DNA to the FANCM repair-support parameter.

Overall, the results obtained demonstrated that the model presented in this paper is both more mechanistic and more biologically informative than the baseline version. It reveals that the dynamic control points lie in the RPA-to-RAD51 exchange step and in FANCM-supported late repair progression, while preserving the known structural relevance of the HR proteins. Indeed, the model integrates structural and dynamical perspectives, showing that proteins identified as important for network topology also become critical in the time-dependent response to parameter perturbation.

4.LIMITATIONS

Despite the improved biological relevance of the proposed model there are two main limitations.

Firstly, many kinetic constants and molecular concentrations involved in the HR are not known experimentally and so the simulations were performed using reasonable parameter ranges and arbitrary units rather than calibrated in vivo values.

Secondly, the mechanistic rules remain simplified compared to the real multistep biochemical behaviour of HR proteins, even though the extension of the RAD51 loading module. For this reason, the project should be interpreted as an exploratory and hypothesis -generating framework rather than a quantitative simulator.

The model is highly useful for identifying influential control points, comparing alternative mechanisms and using it as a baseline for future experimental work, but it should not be used to make exact biological predictions about cellular timescales or concentrations.

5.BIBLIOGRAPHY

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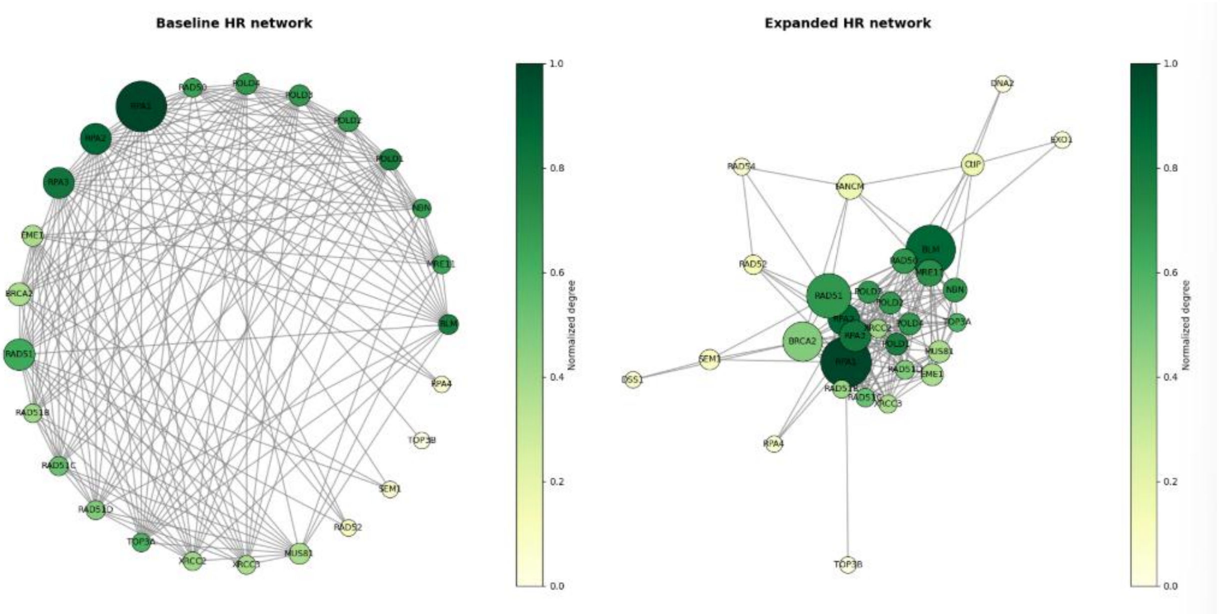

Figure 7 | Comparison between the baseline and expanded homologous recombination.